

Assignment 2 Report

CSE 5525 - Spring 2026 - Transformer Language Modeling

Qifan Wen
wen.679@osu.edu

1 Part 0: Letter Counting BEFOREAFTER Task (Ungraded)

1.1 Hyperparameters and Structure

Parameter	Value	Parameter	Value
vocab_size	27	attention_type	standard
d_model	64	batch_size	32
d_internal	64	lr	0.001
num_layers	1	epochs	30
num_heads	1	layer_norm	pre
causal_mask	False	optimizer	Adam
pos_encoding	False	early_stop	patience=5

Table 1: Part 0 Hyperparameters (BEFOREAFTER scenario)

1.2 Results and Analysis

Metric	Value
Best Dev Accuracy	99.77%
Final Train Accuracy	99.58%
Epochs to Convergence	20 (early stopping)
Best Checkpoint Epoch	15

Table 2: Part 0 Results

2 Part 1: Letter Counting BEFORE Task (35 points)

2.1 Hyperparameters and Structure

Scenario	attn_type	pos_enc	causal	batch
1-layer standard	standard	True	True	32
1-layer relative	relative_pos	True	True	32
1-layer ALiBi	alibi	False	True	32
3-layer standard	standard	True	True	32
4-layer standard	standard	True	True	32

Table 3: Part 1 Hyperparameters (vocab_size=27, num_positions=20, num_classes=3, d_model=64, d_internal=64, num_heads=1, lr=0.001, epochs=30, layer_norm=pre, optimizer=Adam, early_stop=patience=5 for all)

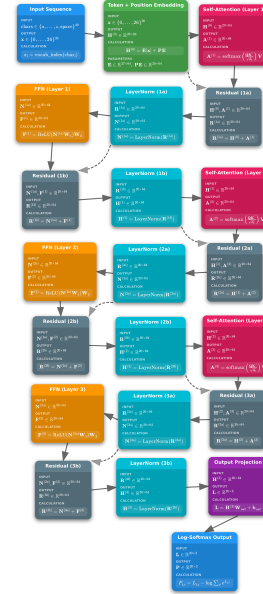


Figure 1: Transformer Architecture

2.2 Results and Analysis

Scenario	Best Dev Acc	Status
1-layer standard	98.93%	Early stop (21)
1-layer relative_position	99.99%	Early stop (15)
1-layer ALiBi	82.95%	Full 30 epochs
3-layer standard	99.93%	Early stop (12)
4-layer standard	99.92%	Early stop (10)

Table 4: Part 1 Results

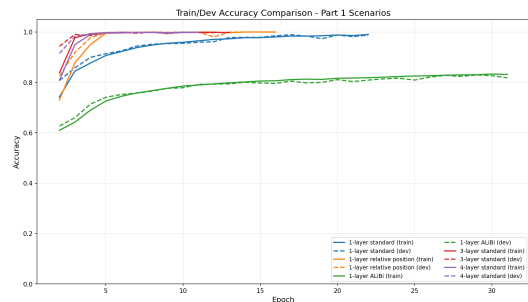


Figure 2: Accuracy Comparison

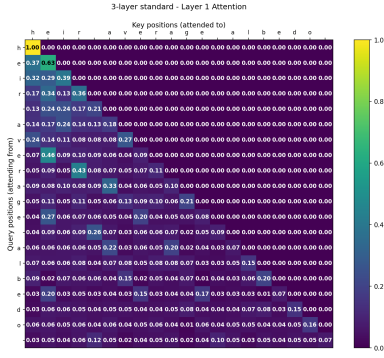


Figure 3: Attention Heatmap

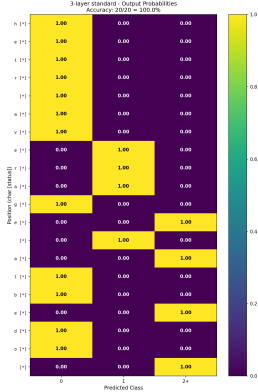


Figure 4: Probability Heatmap

The 1-layer relative_position model achieves the best accuracy (99.99%), outperforming even deeper models, because the BEFORE task is fundamentally about counting occurrences at relative positions and learned relative position embeddings directly encode these positional relationships. Increasing depth from 1 to 3-4 layers significantly improves standard attention (98.93% to 99.93%). The ALiBi attention variant performs poorly (82.95%) because its fixed linear biases penalize distant tokens, but the BEFORE task requires counting all preceding occurrences equally.

3 Part 2: Language Modeling (35 points)

3.1 Hyperparameters and Structure

Scenario	attn_type	pos_enc	early_stop
standard	standard	True	patience=5
relative_position	relative_pos	True	patience=5
ALiBi	alibi	False	patience=5

Table 5: Part 2 Hyperparameters (vocab_size=27, num_positions=128, context_size=64, d_model=128, d_internal=128, layers=2, heads=4, dropout=0.1, batch=64, lr=0.001, epochs=20, layer_norm=pre, optimizer=Adam for all)

The model uses token embeddings, positional encoding (learned or ALiBi), two Transformer layers

with 4-head causal self-attention, and an output projection to vocabulary size.

3.2 Results and Analysis

Scenario	Best Dev PPL	Epochs	Status
standard	9.80	4	Early stop (9)
relative	4.99	19	Full 20 epochs
ALiBi	6.08	20	Full 20 epochs

Table 6: Part 2 Results

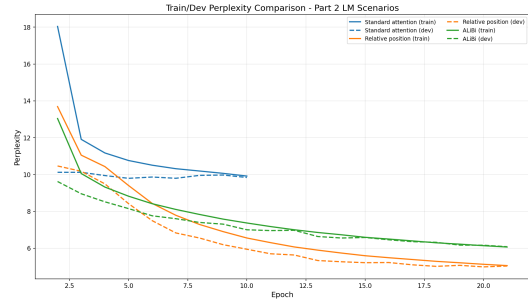


Figure 5: Perplexity Comparison

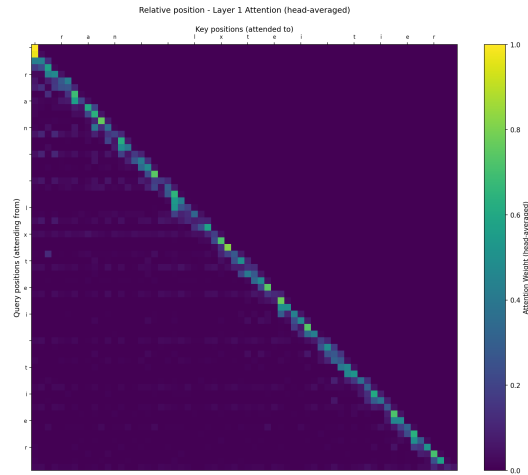


Figure 6: Attention Heatmap

Relative position attention achieves the best perplexity (4.99), improving throughout training and reaching convergence at epoch 19, as the learned relative position embeddings can capture character-level patterns that depend on relative distances. ALiBi outperforms standard attention (6.08 vs 9.80) because the model benefits from different heads having different attention spans. For training dynamics, standard attention shows quick initial learning but early plateau triggering early stopping, ALiBi shows steady improvement throughout training, and relative position shows continuous improvement with the steepest learning curve.